Parameters Extraction for Equivalent Circuit Model Based on Artificial Intelligence

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***Abstract*—In this paper, a novel parameter extraction method for equivalent circuit based on artificial intelligence (AI) technique is proposed. The Y-parameters calculated by UltraEM®, which is a high efficient electromagnetic simulation tool, are used as training data to establish corresponding AI model. With this AI model, the parameters of the equivalent circuit can be efficiently obtained by optimization method. An example of a spiral inductor is given to demonstrate the extraction procedure and validate the results.**

Keywords—parameter extraction, artificial intelligence, optimization method, spiral inductor

I. INTRODUCTION

In radio frequency (RF) integrated circuits, due to parasitic and coupling effects of RF devices at high frequencies, the equivalent circuits of RF devices become more and more complex [1]. However equivalent circuits can give a physical insight of the designed structures. Traditional parameter extraction methods for equivalent circuits based on quasi-static state are not accurate at high frequency. Full wave simulation tools are required for the parameter extraction. Many fast algorithms based on integral equation have been proposed for the accurate and efficient electromagnetic simulation [2], [3], full-wave simulation still requires a certain amount of computational consumption. On the other hand, spiral inductors have been widely used in low noise amplifiers, power amplifiers, voltage controlled oscillators and many other RF devices in silicon-based RF integrated circuits (ICs) [4]. It is required to obtain the equivalent circuit with the corresponding circuit parameters for the physical analysis.

In this paper, a parameter exaction method based on an artificial intelligence (AI) technique, multilayer neural network is proposed. The electromagnetic results of certain frequencies calculated by an efficient full-wave simulation tool for ICs, namely UltraEM® [5], are used as training and testing data for the generation of AI model. Once the AI model is established, the electromagnetic results at any frequencies can be obtained directly from this model. The parameters of equivalent circuits are finally extracted based on the genetic algorithm (GA) [6].

II. PARAMETERS EXTRACTION

The flow of AI model is described in the Fig.1. a. First, Y-parameters at certain frequency points are calculated by UltraEM®, then the data of Y-parameters are normalized. The normalized data is split into training data set and testing data set. It is found that the utilization of 70% data as training data and 30% data as testing data is the optimized combination. The Adam optimization algorithm is used to minimize the loss function *mse(x)*, it can be expressed as:

$$mse(y, y') = \sum_{i=1}^{n} (y_i - y_i^{'})^2 / n \quad (1)$$

where i is the i-th sample, y_i is the real value corresponding to the input term x_i, y_i' is the predicted value corresponding to the input term x_i calculated by neural network, n is the number of samples. After the loss function satisfies the requirement, the AI model is established. As a result, the Y-parameter at any frequency can be obtained through the model.

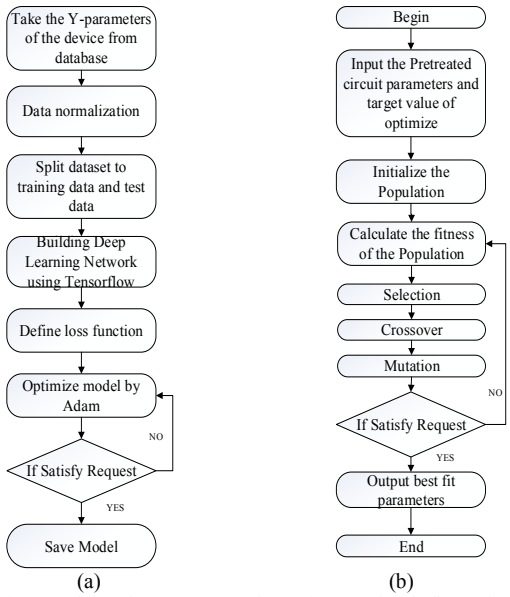


Fig.1 Flows of AI model and GA (a) procedure of AI model (b)flow of GA

A simple GA algorithm is used to optimize the parameters of equivalent circuits. The flow of GA is described in Fig.1. b. First, input some pretreated circuit parameter values as initial population and take the Y-parameters obtained by UltraEM® and AI model as optimization target. The fitness function *fit(x)* defined as:

$$fit(Y, Y') = 1/|Y - Y'| \quad (2)$$

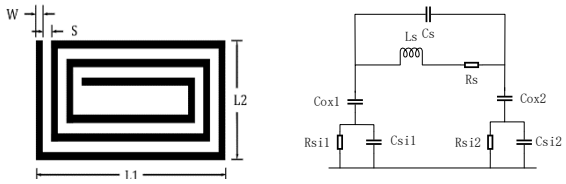
where Y is the optimized Y-parameters obtained by simulating the circuit with the optimized circuit parameters using circuit simulation tool, e.g. Ngspice [7], Y' is the target values. population are selected, crossed and mutated to obtain the individual with the highest fitness function. The error function is defined as the difference between the Y-parameter of the individual with the highest fitness and the optimization

This work was supported in part by the National Key R&D Program of China under Grant 2017YFB0203500, in part by the National Natural Science Foundation of China under Grants 61601163, 61331007, 615040033 and 61411136003.

target value. If the value of error function is smaller than the threshold, it indicates the circuit model matches AI model at a frequency band.

III. Example of a Spiral Inductor

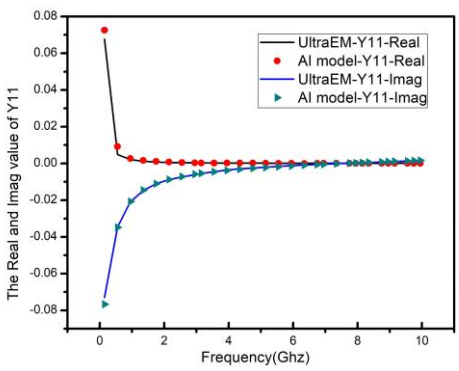
To validate the equivalent circuit with the circuit parameters exacted by the proposed method, a planar spiral inductor shown in Fig.2 has been tested. The inductor is fabricated on a substrate with dielectric constant of 11.9 and thickness of 750 microns. And the thickness and metal conductivity for metal layer are 2 microns and 2.8×10^{7}S/m, respectively.



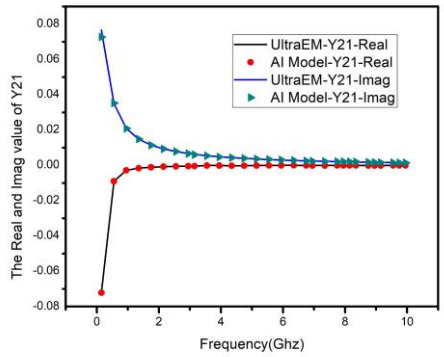
(a) (b)

Fig.2. The planar spiral inductor. (a) geometric structure of the inductor. (b) equivalent circuit for the inductor. The parameters are as the follows (unit: microns) L1=L2=300, W=13, S=7.

At the frequencies of 0.1Ghz to 10Ghz, Y-parameter at 100 frequencies is computed by electromagnetic simulation. Y-parameters at 70 frequencies is used as training data, the remaining data is used for test data. Two hidden layers are set in the input and output layers of the neural network, there are 100 neurons in each hidden layer, the learning rate is 0.01. At the 72 iterations, the value of loss function is 1.635e-5. The Y-parameters generated by UltraEM® and the testing data which is calculated by AI model are shown in Fig.3. The Y parameters results in the frequency range obtained by two method are very close. This shows that the Y-parameter generated by AI model is accurate.



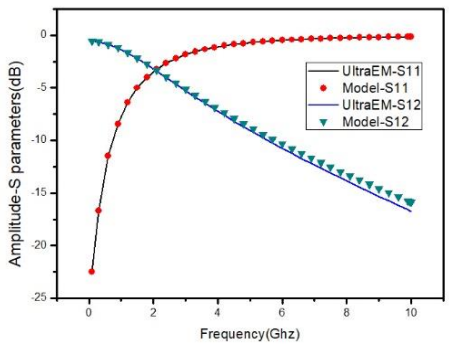
(a)



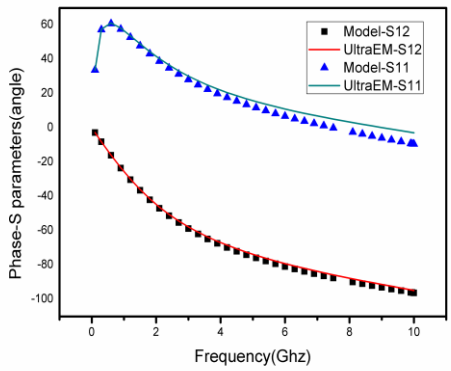
(b)

Fig. 3. Comparison of Y-parameter obtained by UltraEM® and AI model. (a) Comparison of Y11 parameter. (b) Comparison of Y21 parameter.

The extracted parameters optimized by GA are shown in TABLE I. The two-port S-parameters of the circuit model of the planar spiral inductor are generated using SPICE. Two-port S-parameters simulated by UltraEM® and the SPICE are shown in Fig 4.



(a)



(b)

Fig. 4. Comparison of S-parameter results between UltraEM® simulation and circuit model simulation. (a) amplitude comparison of S-parameter. (b) phase comparison of S-parameter

TABLE I

	Circuit components				
The value	Rs(ohm)	Ls(nh)	Cs (fF)	Cox1(pF)	Csi1(pf)
	6.6479	7.7958	6.5243	0.565	6.8901
	Rsi1(ohm)	Cox2(pf)	Csi2(pf)	Rsi2(ohm)	
	5.045	0.363	2.3954	6.3366	

The results of model simulation are in good agreement with the UltraEM® simulation, this shows that the circuit parameters exacted by the method based on AI is accurate, and the method is feasible for parameters exaction.

IV. CONCLUSION

In this paper, a new approach based on AI for parameters exaction is presented. AI model is used to obtain the electromagnetic results at the required frequencies, and the circuit parameters can be generated by this AI model and GA method. A planar spiral inductor is tested, the results demonstrate the accuracy of the method.

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